SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I P. HARINI (192521402) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Harini

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	2	1·8	1·8
Concept Mapping Structure	25%	2·5	2·25	2·25
Linking & Relationships	20%	2	1·8	1·8
Application & Analysis	20%	2	1·8	1·8
Explanation & Clarity	15%	1·5	1·35	1·35

Total Marks 27 **/30**

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:
DataLink→MACAddress→LocalDelivery
Network → IP Address → Global Delivery
Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:
ErrorControl
→DataLinkLayer→ErrorDetection
→ Transport Layer → Error Correction
Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:
Physical ✔ → Data Link ✔ → Network ✖ → Transport ✖ → Application ✖
Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

27/30

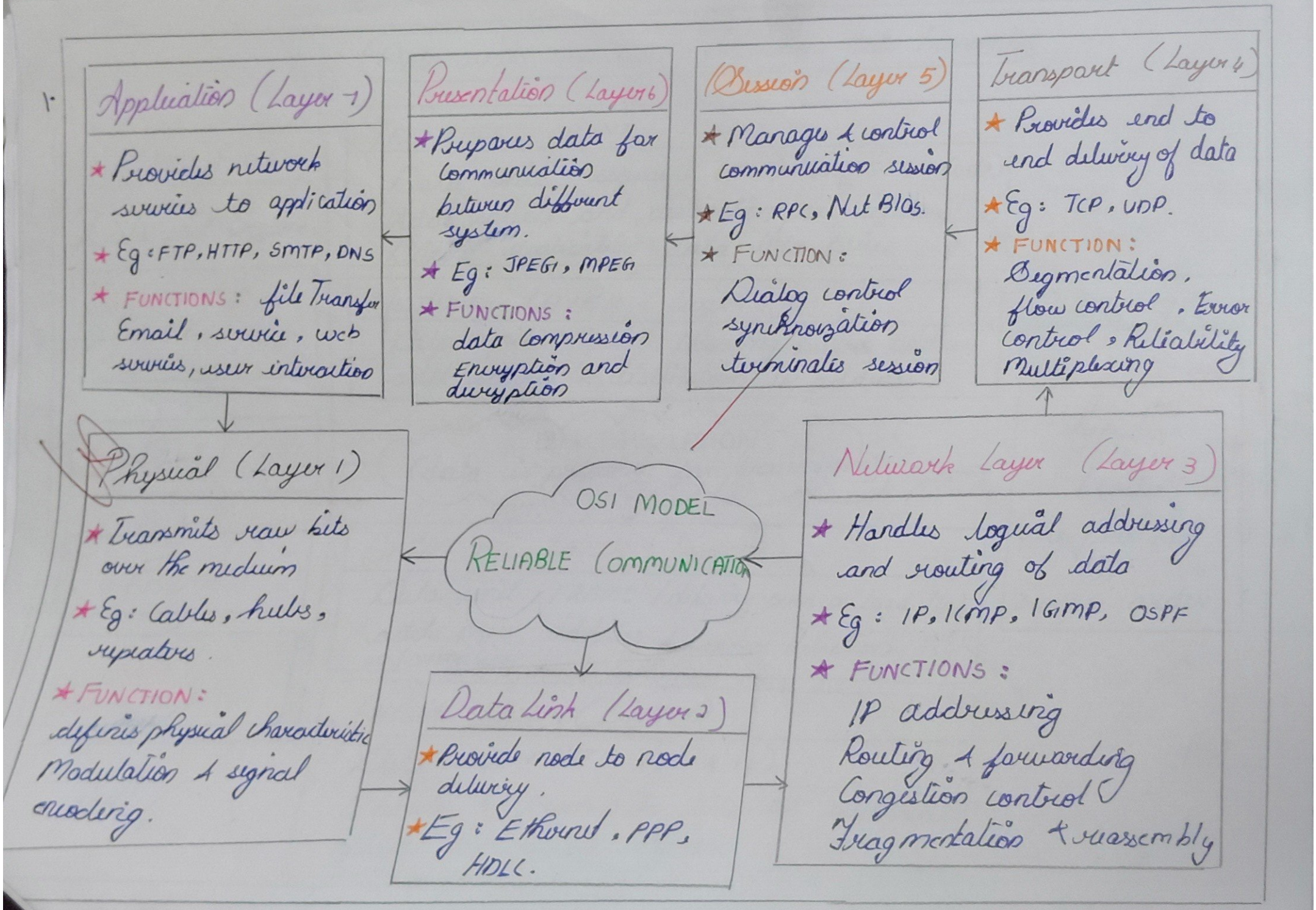
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REG : 192521402

COURSE : Computer Networks

CODE : CSA0710

ASSESMENT: CO1AT2


OSI MODEL
RELIABLE COMMUNICATION
Application (Layer 7)
* Provides network service to application
* Eg: FTP, HTTP, SMTP, DNS
* FUNCTIONS: file Transfer Email, service, web service, user interaction
Presentation (Layer 6)
* Prepares data for communication between different system.
* Eg: JPEG, MPEG
* FUNCTIONS: data compression Encryption and decryption
Session (Layer 5)
* Manages & control communication session
* Eg: RPC, Net BIOS.
* FUNCTION: Dialog control synchronization terminates session
Transport (Layer 4)
* Provides end to end delivery of data
* Eg: TCP, UDP.
* FUNCTION: Segmentation, flow control, Error control, Reliability Multiplexing
Physical (Layer 1)
* Transmits raw bits over the medium
* Eg: Cables, hubs, repeaters.
* FUNCTION: defines physical characteristic Modulation & signal encoding.
Data Link (Layer 2)
* Provide node to node delivery.
* Eg: Ethernet, PPP, HDLC.
Network Layer (Layer 3)
* Handles logical addressing and routing of data
* Eg: IP, ICMP, IGMP, OSPF
* FUNCTIONS:
IP addressing
Routing & forwarding
Congestion control
Fragmentation & reassembly

Data units and OSI Layers - how data Transmitted

TRANSPORT LAYER (Layer 4)

Data unit : segment (TCP) / DATAGRAM (UDP)
Adds source and destination port information
Divides application data into pieces

NETWORK LAYER (Layer 3)

Data unit : Packet / Routing between address
add source & destination IP address

ENCAPSULATION

(data is prepared for transmission)

DATA LINK LAYER (Layer 2)

Data unit : FRAME / delivery over a load link
adds MAC address & error detection information

PHYSICAL LAYER (Layer 1)

Data unit : BITS (0s & 1s)
Encodes & transmits bits as electrical optical or radio signals

SENDER SIDE

Data unit Sequence :

Segment / Data -gram (TCP / UDP)

↓

Packet

↓

Frame

↓

Bits (0s & 1s)

Receiver Side

Decapsulation

headers
+
trailers
are processed
in
reverse order

3.

MAC & IP ADDRESS - Complete data Delivery

Sender
Data + destination IP

Local Network
Data link layer MAC address delivery frame on local Link

Router
Reads destination IP selects next route replaces frame for next link

REMOTE Network
IP address guides packet across network MAC changes each link

Receiver
destination IP matches frame accepted data delivery

MAC VS IP - QUICK COMPARISON

Address type	OSI Layer	Delivery scope	key features
MAC address	Layer 2	Local delivery	Changes at each link
IP address	Layer 3	Global delivery	Identifies source / destination across network